

A one-step weight shift proves the conjectured interpolation identity

Candidate full answer to Remark 4.9 of arXiv:2507.13478

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Status: candidate full affirmative answer. For every parameter allowed in Remark 4.9, the two complex interpolation spaces in equation (4.3) coincide with equivalent norms. Multiplication by the normal coordinate moves the problem to a Muckenhoupt weight. There the two endpoint boundary systems differ only by a second-normal-trace condition, which Roodenburg's arbitrary-smoothness interpolation theorem makes invisible exactly below the conjectured threshold.

1 The question

Let $p \in (1, \infty)$, let X be a UMD Banach space, and put $w_\gamma(x) = x_1^\gamma$ on the half-space $\mathbb{R}_+^d = \{x_1 > 0\}$. Lindemulder, Lorist, Roodenburg, and Veraar [1], Remark 4.9, conjecture that for

$$p - 1 < \gamma < 2p - 1, \quad 0 < \theta < \frac{1}{3} \left(1 + \frac{\gamma + 1}{p} \right),$$

one has

$$[L^p(w_\gamma; X), W_{\text{Dir}}^{3,p}(w_\gamma; X)]_\theta = [L^p(w_\gamma; X), W_0^{3,p}(w_\gamma; X)]_\theta. \quad (1)$$

Here and below all spaces are on $\mathbb{R}_+^d$, and equality means equality of Banach spaces with equivalent norms.

proving the proposition. □

Remark 4.9. We conjecture that, e.g., in the case $k = k_0 = 0$, there is actually the equality of complex interpolation spaces

$$[L^p(\mathbb{R}_+^d, w_\gamma; X), W_{\text{Dir}}^{3,p}(\mathbb{R}_+^d, w_\gamma; X)]_\theta = [L^p(\mathbb{R}_+^d, w_\gamma; X), W_0^{3,p}(\mathbb{R}_+^d, w_\gamma; X)]_\theta \quad (4.3)$$

for all $\theta \in (0, \frac{1}{3}(1 + \frac{\gamma+1}{p}))$, which is suggested by results on interpolation with boundary conditions as studied in [71, 82]. However, at the moment, the case $\gamma \in (p - 1, 2p - 1)$ of (4.3) for the parameter range $\theta \in (\frac{2}{3} + \varepsilon, \frac{1}{3}(1 + \frac{\gamma+1}{p}))$ is an interesting open problem that seems to require a novel approach to interpolation with boundary conditions.

Figure 1: Source evidence from PDF page 21 of arXiv:2507.13478, Remark 4.9. The paper states the identity and singles out the remaining upper parameter range as an interesting open problem.

For the present range of γ , the trace characterisations in the source reduce to

$$\begin{aligned} W_{\text{Dir}}^{3,p}(w_\gamma; X) &= \{f \in W^{3,p}(w_\gamma; X) : \text{Tr}_0 f = 0\}, \\ W_0^{3,p}(w_\gamma; X) &= \{f \in W^{3,p}(w_\gamma; X) : \text{Tr}_0 f = \text{Tr}_1 f = 0\}. \end{aligned}$$

Only normal traces are displayed: the tangential first-order traces vanish automatically when $\text{Tr}_0 f = 0$. We write $\text{Tr}_j f = (\partial_1^j f)|_{x_1=0}$.

2 Proof intuition

The missing interval is difficult only because w_γ lies outside the Muckenhoupt range, where fractional Sobolev interpolation is not intrinsically identified. Multiplication by x_1 changes w_γ into $w_{\gamma-p}$. It also shifts boundary jets by one order: Dirichlet data become two zero normal traces, and full-zero data become three.

The latter endpoint shift is a known isomorphism. For the former, the only issue is surjectivity. Given a lower-weight function whose first two traces vanish, lift its second trace at the original weight, multiply that lift by x_1 , and subtract. The remainder has all three lower-weight traces zero and can be divided by x_1 using the full-zero isomorphism.

After this endpoint identification, both couples live at the Muckenhoupt weight. Roodenburg's theorem says that an order- m boundary condition appears at intermediate smoothness s only when $s > m + (\beta + 1)/p$. Hence the two systems agree until the order-two trace appears. That appearance occurs at exactly the source's upper threshold.

3 The endpoint weight shift

Set

$$\beta := \gamma - p, \quad a := \frac{\beta + 1}{p} = \frac{\gamma + 1}{p} - 1.$$

Then $-1 < \beta < p - 1$ and $0 < a < 1$. Define the lower-weight trace kernels

$$\begin{aligned} E_{01} &:= \{g \in W^{3,p}(w_\beta; X) : \text{Tr}_0 g = \text{Tr}_1 g = 0\}, \\ E_{012} &:= \{g \in W^{3,p}(w_\beta; X) : \text{Tr}_0 g = \text{Tr}_1 g = \text{Tr}_2 g = 0\}. \end{aligned}$$

The trace characterisation gives $E_{012} = W_0^{3,p}(w_\beta; X)$.

Lemma 1 (Endpoint shift). *For $(Mf)(x) = x_1 f(x)$, the following are Banach-space isomorphisms:*

$$\begin{aligned} M &: L^p(w_\gamma; X) \longrightarrow L^p(w_\beta; X), \\ M &: W_{\text{Dir}}^{3,p}(w_\gamma; X) \longrightarrow E_{01}, \\ M &: W_0^{3,p}(w_\gamma; X) \longrightarrow E_{012}. \end{aligned}$$

Proof. The first map is an isometry because

$$\|Mf\|_{L^p(w_\beta)}^p = \int_{\mathbb{R}_+^d} |x_1 f(x)|^p x_1^{\gamma-p} dx = \|f\|_{L^p(w_\gamma)}^p.$$

The third map is precisely the multiplication isomorphism on full-zero weighted Sobolev spaces [3, Lemma 3.6(ii)].

For the second map, boundedness into $W^{3,p}(w_\beta; X)$ follows from [3, Lemma 3.6(i)]. The product rule gives, initially on smooth functions and then by trace continuity,

$$\text{Tr}_0(Mf) = 0, \quad \text{Tr}_1(Mf) = \text{Tr}_0 f, \quad \text{Tr}_2(Mf) = 2 \text{Tr}_1 f. \quad (2)$$

Thus M maps the Dirichlet endpoint into E_{01} .

It remains to prove surjectivity with a bounded inverse. Let $g \in E_{01}$ and put $q = \text{Tr}_2 g$. The lower-weight trace theorem gives

$$q \in B_{p,p}^{3-2-a}(\mathbb{R}^{d-1}; X) = B_{p,p}^{2-(\gamma+1)/p}(\mathbb{R}^{d-1}; X).$$

At the original weight, the universal right inverse for the first normal trace [2, Theorem 4.6] therefore supplies

$$e := \frac{1}{2} \text{ext}_1 q \in W^{3,p}(w_\gamma; X), \quad \text{Tr}_0 e = 0, \quad \text{Tr}_1 e = q/2,$$

with $\|e\|_{W^{3,p}(w_\gamma)} \lesssim \|q\|_{B_{p,p}^{1-a}}$. By (2), the function $h := g - Me$ satisfies $\text{Tr}_0 h = \text{Tr}_1 h = \text{Tr}_2 h = 0$, so $h \in E_{012}$. The third isomorphism yields a unique $f_0 \in W_0^{3,p}(w_\gamma; X)$ such that $Mf_0 = h$. Consequently

$$f := f_0 + e \in W_{\text{Dir}}^{3,p}(w_\gamma; X), \quad Mf = g.$$

The continuity of Tr_2 , of ext_1 , of M , and of the inverse on the full-zero spaces gives $\|f\|_{W^{3,p}(w_\gamma)} \lesssim \|g\|_{W^{3,p}(w_\beta)}$. This proves the asserted isomorphism. $\square$

4 The interpolation identity

Theorem 2. *Let $p \in (1, \infty)$, $p - 1 < \gamma < 2p - 1$, and let X be a UMD Banach space. Then (1) holds for every*

$$0 < \theta < \frac{1}{3} \left(1 + \frac{\gamma + 1}{p} \right).$$

Thus the conjecture in Remark 4.9 of the source is true.

Proof. By Lemma 1 and functoriality of complex interpolation, applying M to the left and right sides of (1) gives respectively

$$[L^p(w_\beta; X), E_{01}]_\theta, \quad [L^p(w_\beta; X), E_{012}]_\theta. \quad (3)$$

Because $-1 < \beta < p - 1$ and X is UMD, integer-order weighted Bessel-potential and Sobolev spaces coincide. We may therefore apply Roodenburg's complex interpolation theorem for weighted Bessel-potential spaces with normal boundary systems [2, Theorem 6.4], with $s_0 = 0$, $s_1 = 3$, and $s_\theta = 3\theta$. Its lower-index hypothesis holds because $0 > -1 + a$.

First suppose

$$3\theta \notin \{a, 1 + a, 2 + a\}.$$

The theorem identifies the first space in (3) with the lower-weight Bessel-potential space of smoothness 3θ carrying the trace conditions among Tr_0, Tr_1 for which

$$j + a < 3\theta.$$

It identifies the second space in exactly the same way, except that Tr_2 is also available at the top endpoint. But the assumed parameter range is

$$3\theta < 1 + \frac{\gamma + 1}{p} = 2 + a.$$

Hence the condition $\text{Tr}_2 = 0$ has not yet appeared. The two intermediate spaces, with their norms, are therefore identical.

Only the critical values $3\theta = a$ and $3\theta = 1 + a$ remain inside the open interval. Fix either such θ_0 . Choose noncritical $0 < r < \theta_0 < s < (2 + a)/3$ and $\lambda \in (0, 1)$ with $\theta_0 = (1 - \lambda)r + \lambda s$. If $I_{01}(t)$ and $I_{012}(t)$ denote the two shifted interpolation scales, the preceding paragraph gives $I_{01}(r) = I_{012}(r)$ and $I_{01}(s) = I_{012}(s)$. Exact complex reiteration gives

$$I_{01}(\theta_0) = [I_{01}(r), I_{01}(s)]_\lambda = [I_{012}(r), I_{012}(s)]_\lambda = I_{012}(\theta_0).$$

The couples are regular: smooth functions supported in the open half-space belong to both top endpoints and are dense in the common L^p endpoint. Thus reiteration applies. Pulling the equality back through M completes the proof for all θ in the claimed interval. $\square$

5 Scope, novelty, and review priorities

The theorem answers exactly the half-space interpolation conjecture displayed in Remark 4.9. It does not address the source's separate questions about negative smoothness, Lipschitz domains, or the unshifted Laplacian at $\mu = 0$. No endpoint assertion is made at $3\theta = 1 + (\gamma + 1)/p$, where the second normal trace reaches its critical smoothness.

The local run indexes were searched by arXiv id, title, and core terms. External exact-phrase, title/citation, author-page, and arXiv searches through 12 August 2026 found no later paper explicitly resolving equation (4.3). The source already cites both ingredients used here, but does not record the endpoint identification in Lemma 1. Novelty is therefore plausible, not certified. Human review should focus first on the surjectivity argument in Lemma 1, and second on reiteration at the two critical trace indices.

References

- [1] N. Lindemulder, E. Lorist, F. B. Roodenburg, and M. C. Veraar, *Functional calculus on weighted Sobolev spaces for the Laplacian on rough domains*, J. Differential Equations 454 (2026), 113884; arXiv:2507.13478.
- [2] F. B. Roodenburg, *Complex interpolation of power-weighted Sobolev spaces with boundary conditions*, arXiv:2503.14636, current version February 2026.
- [3] N. Lindemulder, E. Lorist, F. B. Roodenburg, and M. C. Veraar, *Functional calculus on weighted Sobolev spaces for the Laplacian on the half-space*, J. Funct. Anal. 289 (2025), 110985; arXiv:2406.03297.